

PROBLEM SHEET 3: SOLUTIONS & FEEDBACK

1. (a) For small u , the interaction term behaves like auv , i.e. like standard Lotka–Volterra predator–prey. For large u , the interaction term behaves like av , i.e. like prey consumption rate is only dependent on the predator population, rather than the opportunities for consumption between predator and prey. Just like in the spruce budworm model, this is saying that the *predators get full*.

Feedback

The errors/mark losses made here fell into two classes:

- Not being precise enough about the behaviour mathematically – the crucial point is that the limiting behaviour is very different from the standard uv term as u gets very large.
- Not offering a rough interpretation (saturation).

When we ask you to describe terms we are always looking for mathematical descriptions, often what it does for small u and large u . When it comes to interpretations, it's normally best to compare to the standard Lotka–Volterra-type uv in which the interaction is linearly dependent on population size: this implies the predators can always eat/out compete a certain percentage of the population no matter how big it is!

- (b) Looking for equilibria, we say

$$0 = u \left(1 - \frac{u}{k} \right) - \frac{auv}{1+u}, \quad (1a)$$

$$0 = -bv + \frac{auv}{1+u}. \quad (1b)$$

Looking first at eq. (1b), we have

$$0 = v \left(-b + \frac{au}{1+u} \right) \implies v = 0, \quad b = \frac{au}{1+u}. \quad (2)$$

Plugging the first solution, $v = 0$, into eq. (1a), we get $u = 0$ or $u = k$, so we have found $(0, 0)$ and $(k, 0)$.

The second solution rearranges to

$$u = \frac{b}{a-b}. \quad (3)$$

Rearranging eq. (1a) for v and substituting this solution for u in, we get

$$v = \frac{1+u}{a} \left[1 - \frac{u}{k} \right] = \dots = \frac{k(a-b) - b}{k(a-b)^2}, \quad (4)$$

and so our final equilibrium is

$$(u^*, v^*) = \left(\frac{b}{a-b}, \frac{k(a-b) - b}{k(a-b)^2} \right). \quad (5)$$

This is permissible if $a > b$ and $k(a-b) \geq b$.

 Feedback

Up to a few errors this was fine; nothing obviously systematic.

(c) If $a = 2$ and $b = 1$ then

$$(u^*, v^*) = \left(1, \frac{k-1}{k}\right), \quad (6)$$

and this is permissible if $k \geq 1$.

 Feedback

Generally fine!

(d) Under this choice for a and b , the Jacobian is

$$\mathbf{J} = \begin{pmatrix} 1 - \frac{2u}{k} - \frac{2v}{(1+u)^2} & -\frac{2u}{1+u} \\ \frac{2v}{(1+u)^2} & -1 + \frac{2u}{1+u} \end{pmatrix}. \quad (7)$$

Evaluated at each equilibrium, we have:

- at $(0, 0)$:

$$\mathbf{J}(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \implies \lambda = 1, -1. \quad (8)$$

So this is unstable (a saddle point).

- at $(k, 0)$:

$$\mathbf{J}(k, 0) = \begin{pmatrix} -1 & -\frac{k}{1+k} \\ 0 & -1 + \frac{2k}{1+k} \end{pmatrix} \implies \lambda = -1, -1 + \frac{2k}{1+k}. \quad (9)$$

This is unstable if the second eigenvalue is positive, i.e. if

$$-1 + \frac{2k}{1+k} > 0 \implies k > 1. \quad (10)$$

So it is stable for $0 < k < 1$ (recall k is a positive constant), and unstable for $k > 1$.

- at $(u^*, v^*) = (1, (k-1)/k)$:

$$\mathbf{J}(u^*, v^*) = \begin{pmatrix} 1 - \frac{2}{k} - \frac{1}{2} \left(\frac{k-1}{k}\right) & -1 \\ \frac{1}{2} \left(\frac{k-1}{k}\right) & 0 \end{pmatrix} = \begin{pmatrix} \frac{k-3}{2k} & -1 \\ \frac{k-1}{2k} & 0 \end{pmatrix}. \quad (11)$$

Following the hint, we have

$$\det \mathbf{J} = \frac{k-1}{2k}, \quad \text{tr } \mathbf{J} = \frac{k-3}{2k}, \quad (\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = \frac{-7k^2 + 2k + 9}{4k^2}. \quad (12)$$

Now we can see the determinant and trace change signs at $k = 1$ and $k = 3$, so let's think about the regions either side and between these points:

- If $0 < k < 1$: the equilibrium is not permissible.
- If $1 < k < 3$: $\det \mathbf{J} > 0$, $\text{tr } \mathbf{J} < 0$, so the equilibrium is stable.
- If $k > 3$: $\det \mathbf{J} > 0$, $\text{tr } \mathbf{J} > 0$, so the equilibrium is unstable.

We can put this information all together:

u_0	v_0	permissible?	stable?	unstable?
0	0	always	—	always
k	0	always	$0 < k < 1$	$k > 1$
1	$\frac{k-1}{k}$	$k > 1$	$1 < k < 3$	$k > 3$

Feedback

Everyone knows what to do with stability analysis now. There were two common errors:

- Assuming $k < 1$ from the previous result applied to the $(k, 0)$ equilibrium, but it was only for coexistence. Take care and review your answers, the clue here is that $(k, 0)$ is clearly sensible if $k \geq 0$.
- A lot of people looked at the discriminant of the quadratic for λ for stability finding, essentially when they switched from node-like behaviour to spiral behaviour. This is *not* important just for the sake of stability. Just look at $\text{tr } \mathbf{J}$ and $\det \mathbf{J}$, in which case the answer is very simple.

There is an important principle here: *never do more than is needed* in an exam. You only have a finite amount of time, so just answer what is asked and no more. If it says *classify* the equilibria, you need to know if you have complex behaviour or not; if it asks for stability you don't.

- (e) Look again at the table above. Ignoring the complete extinction equilibrium, we have

$$(u_0, v_0) = (k, 0) \quad \text{and} \quad \left(1, \frac{k-1}{k}\right). \quad (13)$$

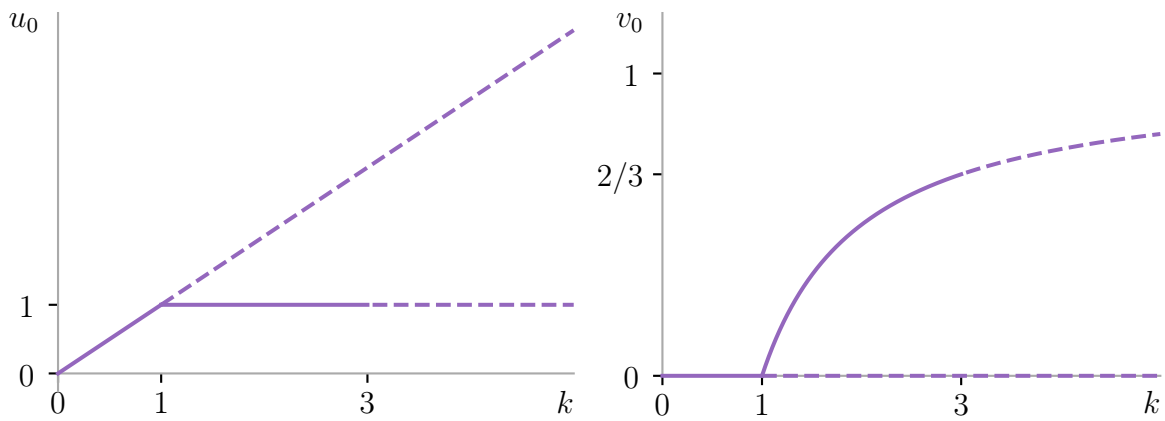
So for the u_0 bifurcation diagram we have:

- $u_0 = k$, permissible for all k and stable for $0 < k < 1$, unstable otherwise,
- $u_0 = 1$, permissible only for $k > 1$, stable for $1 < k < 3$, unstable otherwise.

And for the v_0 bifurcation diagram we have:

- $v_0 = 0$, permissible for all k and stable for $0 < k < 1$, unstable otherwise,
- $v_0 = (k-1)/k$, permissible only for $k > 1$, stable for $1 < k < 3$, unstable otherwise.

These look on the graph like:



Feedback

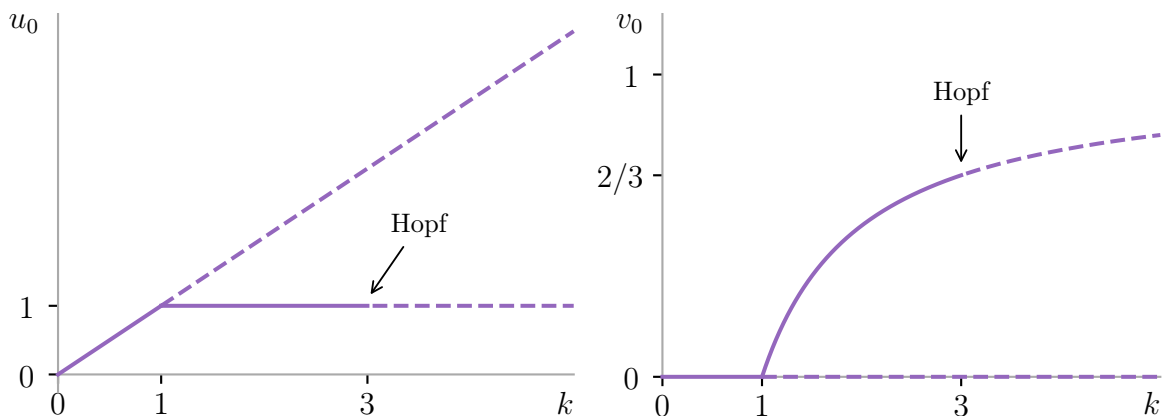
This was a mixed bag. I think some people weren't clear on what a bifurcation diagram is, in which case, just make sure you understand what is above. Common mistakes included drawing the wrong stability on the lines (remember stable equilibria get solid lines; unstable equilibria get dashed lines), drawing impermissible equilibria ($u_0 = 1$ is only permissible for $k = 1$ so don't draw anything for $k < 1$) and just missing equilibria (some people missed the line for $v_0 = 0$).

- (f) The Hopf bifurcation is the point where a stable spiral becomes an unstable spiral (or the other way round).

Looking at the graph, there are two bifurcations where stability changes: one at $k = 1$ and $k = 3$.

At $k = 1$, $(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = 1 > 0$, so the bifurcation there is a transition between nodes, not spirals (look at the $\text{tr}/\det$ table in the notes).

At $k = 3$, $(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J} = -4/3 < 0$, so this is the bifurcation between spirals. Hence this is the Hopf bifurcation and we can mark it on our diagrams:



Feedback

A lot of people said $k = 1$ was the location of a Hopf bifurcation. But a Hopf bifurcation is when the solution changes between a stable *spiral* and an unstable *spiral*: i.e. it's at $k = 3$. (So now the discriminant matters!)

2. (a) We start with the advection–diffusion equation, which in one dimension is

$$\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}[uv] - \frac{\partial J}{\partial x} + f \quad (14)$$

where J is the flux, f a source and v the bulk velocity.

- The question states $v = 0$.
- Fick’s law in one dimension is $J = -D\partial u/\partial x$, with D the diffusion constant.
- Finally, the bacteria death mentioned in the text is clearly a source term $-\lambda u$ (or a sink as it is sometimes called when negative).

Put together, this gives the required equation.

The boundary conditions will be at $x = 0$ and $x = L$. We put the ‘death’ condition at $x = 0$ which requires the population is zero at this point, $u(0, t) = 0$. The density gradient condition requires that $\partial u/\partial x(L, t) = \mu$.

This is the correct sign on μ because Fick’s law states that density flows from higher to lower density. At $x = L$, we want the bacteria to come in (move to the left), so we want more concentration on the right, i.e. a positive gradient.

Feedback

A good number of marks were lost here, almost always for the same reason. Basically you have to start with the *general* advection diffusion model, which in 1D is

$$\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}[uv] - \frac{\partial J}{\partial x} + f. \quad (15)$$

We derived this in class from only assuming mass conservation. In the derivation J is the spreading out (or inward flow) of the density from a volume element of the domain (a line element in 1D). That’s basically what diffusion is: spreading.

Many of you started with

$$\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}[uv] + D\frac{\partial^2 u}{\partial x^2} + f, \quad (16)$$

which implicitly assumed Fickian diffusion. That is quite common, but as we saw in §5.5, there are plenty of models that are nonlinear, with diffusion $D(u)$ having a dependence on u . The point is you have to make a choice as to what J is.

Some also assumed

$$\frac{\partial u}{\partial t} = -v\frac{\partial}{\partial x}[u] + D\frac{\partial^2 u}{\partial x^2} + f, \quad (17)$$

which assumes constant v . Again, that is fine but you have to make that assumption first.

Some of you also got the boundary condition wrong at $x = L$, saying $-\mu$ instead of μ . I suspect most of you didn’t think about it too much and got it right by default, but if the problem were set up reflected, you would need $-\mu$.

- (b) The equilibrium requires that

$$D\frac{d^2 u}{dx^2} - u = 0. \quad (18)$$

We either write down the auxiliary equation in m , or just go straight for exponential solutions, $u = e^{mx}$, giving us

$$Dm^2 - 1 = 0 \implies m = \pm \frac{1}{\sqrt{D}}. \quad (19)$$

So the general solution is

$$u(x) = \tilde{A}e^{-x/\sqrt{D}} + \tilde{B}e^{x/\sqrt{D}}, \quad (20)$$

although the solution in the question suggests we should write it in terms of hyperbolic trig functions,

$$u(x) = A \cosh\left(\frac{x}{\sqrt{D}}\right) + B \sinh\left(\frac{x}{\sqrt{D}}\right). \quad (21)$$

The first boundary condition at $x = 0$ requires $A = 0$ so

$$u(x) = B \sinh\left(\frac{x}{\sqrt{D}}\right). \quad (22)$$

The x -derivative is

$$\frac{du}{dx} = \frac{B}{\sqrt{D}} \cosh\left(\frac{x}{\sqrt{D}}\right), \quad (23)$$

and the second boundary condition reads

$$\frac{B}{\sqrt{D}} \cosh\left(\frac{L}{\sqrt{D}}\right) = \mu \implies B = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})}. \quad (24)$$

So the final solution is

$$u(x) = \frac{\sqrt{D}\mu \sinh(x/\sqrt{D})}{\cosh(L/\sqrt{D})}, \quad (25)$$

i.e.

$$\alpha = \sqrt{D}, \quad \beta = \frac{L}{\sqrt{D}}. \quad (26)$$

Feedback

Done mostly fine. Some of you got the right answer for u but didn't quite get it into the correct form, and hence wrote down the wrong values of α and β . Be careful!

(c) To find the centre of mass we must integrate,

$$\int_0^L xu(x) dx = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})} \int_0^L x \sinh\left(\frac{x}{\sqrt{D}}\right) dx. \quad (27)$$

We use integration by parts to get

$$\int_0^L x \sinh\left(\frac{x}{\sqrt{D}}\right) dx = \sqrt{D} \left[x \cosh\left(\frac{x}{\sqrt{D}}\right) \right]_0^L - \sqrt{D} \int_0^L \cosh\left(\frac{x}{\sqrt{D}}\right) dx \quad (28)$$

$$= \sqrt{D}L \cosh\left(\frac{L}{\sqrt{D}}\right) - D \left[\sinh\left(\frac{x}{\sqrt{D}}\right) \right]_0^L \quad (29)$$

$$= \sqrt{D}L \cosh\left(\frac{L}{\sqrt{D}}\right) - D \sinh\left(\frac{L}{\sqrt{D}}\right). \quad (30)$$

The centre of mass, C_m , is given by

$$C_m = \frac{\int_0^L xu(x) \, dx}{\int_0^L u(x) \, dx}, \quad (31)$$

and

$$\int_0^L u(x) \, dx = \frac{\sqrt{D}\mu}{\cosh(L/\sqrt{D})} \int_0^L \sinh\left(\frac{x}{\sqrt{D}}\right) \, dx \quad (32)$$

$$= \frac{D\mu}{\cosh(L/\sqrt{D})} \left[\cosh\left(\frac{L}{\sqrt{D}}\right) - 1 \right]. \quad (33)$$

So

$$C_m = \frac{L \cosh(L/\sqrt{D}) - \sqrt{D} \sinh(L/\sqrt{D})}{\cosh(L/\sqrt{D}) - 1}, \quad (34)$$

i.e.

$$\alpha = \sqrt{D}, \quad \beta = \frac{L}{\sqrt{D}}, \quad \gamma = L, \quad \delta = 1. \quad (35)$$

Feedback

Mostly this was done fine, with just a few integration errors.

- (d) In this limit $L/\sqrt{D} \gg 1$. Hence $\sinh(L/\sqrt{D}) \approx \cosh(L/\sqrt{D})$ and $\cosh(L/\sqrt{D}) \gg 1$ so $C_m \approx L$ and the whole distribution gets squashed against the boundary at $x = L$.

Feedback

A subtle point here is that we don't need $L \rightarrow \infty$: it could be that D is very small and L is order 1. Then the result of the centre of mass tending to L not ∞ matters. For those of you who put $C_m \rightarrow L - \sqrt{D}$ not L , it's important for the interpretation that the mass of bacteria end up at the source of inflow at $x = L$. The final thing that some people didn't do was to actually interpret the result $C_m \rightarrow L$. Make sure you answer the question!